VF Asia Footwear Laboratory Testing Manual

Test Method for Stitch Tear	Test Method #: VF- B08 Page 1 of 3
Date Issued: May, 2019 Revision #: 1 Reason for revision: Rectify the unit of measure from kg to kgf for material strength.	Created by: Nila Babia Erazo

PURPOSE AND SCOPE: To measure the load required to tear a material which has been perforated by a row of stitching needles along an edge of the test material.

REFERENCES: SATRA TM33

EQUIPMENT AND MATERIAL:

1. Tensile testing machine ,meeting the requirements of ASTM D638
2. Clamping jaw ,preferably rubber-faced
3. Needle holding jig(SATRA STD176)
4. Needles, round point 16*1 size 14.
5. Fabric, leather, synthetic, non-woven and other stitched upper materials (e.g., rubber, plastic, etc.)

SAMPLE PREPARATION:

1. Condition the sample for at least 24 hours at the standard laboratory conditions at 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

2. Die or hand cut test samples in conformance with **Figure A**. Fabrics should be cut such that the yarns are parallel to the top and bottom of the test sample.

Note 1: For woven fabrics, after the test samples are cut, make sure that the bottom of the sample has a “full yarn” across the entire width of the sample. If it does not, use a sharp pair of scissors to cut the sample so that it does. This is particularly important with woven fabrics using heavier yarns.

Note 2: With fabrics, the test samples should be taken such that no two samples contain the same warp and fill yarns.

Test Method for Stitch Tear	Test Method #: VF- B08 Page 2 of 3
Date Issued: May, 2019 Revision #: 1 Reason for revision: Rectify the unit of measure from kg to kgf for material strength.	Created by: Nila Babia Erazo

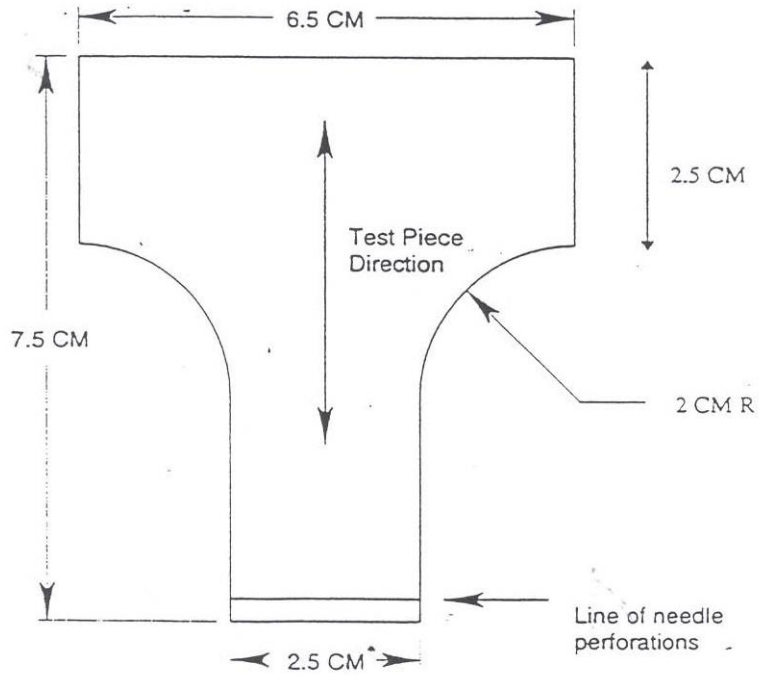


Figure A

REPETITIONS: Test three (3) samples in each direction per material. The length direction corresponds to the following.

a. Fabrics = line of needles is parallel to the end yarns.
b. Leather = line of needles is parallel to the backbone.
c. Synthetic = line of needles is parallel to the machine direction.

TEST PROCEDURE:

Test should be carried out at standard laboratory conditions @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

1. Set **Plate C** of the needle holding jig, such that the margin is 3.0+/-0.1mm.

Note: For Webbings **Plate C** should be set at 6.0+/-0.1m

Test Method for Stitch Tear	Test Method #: VF- B08 Page 3 of 3
Date Issued: May, 2019 **Revision #: 1** **Reason for revision: Rectify the unit of measure from kg to kgf for material strength.**	**Created by: Nila Babia Erazo**

2. Insert the test sample in the jig such that the base butts against the end of **Plate C**, and the sample is aligned centrally between the two end needle holes.
3. Push the 17 needles through the opening in the holding jig and the test sample, such that the tip of the needles passes completely through the test fabric and into the bottom plate.
4. Insert the holding jig in the bottom fixture of the tensile testing machine.
5. Zero and calibrate the tensile testing machine, set the crosshead speed to 100mm/minute.
6. Insert and clamp the free end of the test material centrally in the top jaw of the tensile testing machine, to a depth of 2.0-2.5cm.
7. Operate the tensile testing machine at a crosshead speed of 100mm/minute.
8. Record the maximum load required to tear through the test material to the nearest 0.1kgf.

CALCULATION:

$$\textbf{Stitch tear strength (kgf /cm)} = \frac{L}{2.5}$$

Note: For samples less than 2.5cm divide the load by the actual width of the sample (not 2.5).

REPORT: The average of the three stitch tear measurements to the nearest 0.1kgf/cm